

Tanishq Joshi

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Summary

Software Engineer with **2 years** of experience building high-scale **web applications** and **AI orchestration** systems using **Golang** (Gin) and **Python** (FastAPI). Expert in optimizing backend performance and managing databases, delivering **3x latency improvements** and **\$2M+ in enterprise value**. Proven track record in architecting **RAG pipelines** and event-driven **microservices** on RabbitMQ to drive operational efficiency and scalability in production environments.

Education

Birla Institute of Technology and Science (BITS), Pilani

M.Sc. (Hons.) Mathematics | B.E. (Hons.) Electronics & Communication Engineering

CGPA: 8.33 / 10

November 2020 – July 2025

Experience

Telus Digital

Bengaluru, India

Software Development Engineer I

July 2025 – Present

- Architected a FastAPI orchestration layer for a **RAG-powered AI assistant**, integrating a temporal knowledge graph (Amazon Neptune) with hybrid BM25 + vector search over an event-driven **RabbitMQ** pipeline, boosting rater accuracy by **20%** across **50+ enterprise workflows** and cutting query latency by **30%**.
- Overhauled the **multi-tenant PostgreSQL** data model, migrating from MongoDB to normalized relational schemas with composite indexing and query plan optimization; slashed client onboarding time by **85%** and achieved a **3x reduction in p95 latency**.
- Implemented an event-driven bulk processing pipeline on RabbitMQ with adaptive batching and fault-tolerant retry logic, eliminating N+1 bottlenecks to deliver **2x throughput** across **10K+ daily workflows** and cut API calls by **40%**.
- Shipped a configurable model interaction workflow on FTS with per-workflow endpoint schema config, unlocking **\$2M** in pilot contracts across **4 enterprise clients** and enabling annotator-driven, multi-turn **RLHF** correction loops for enterprise fine-tuning.

Software Development Engineer Intern

December 2024 – June 2025

- Saved **\$50K+** annually in LLM inference and annotation costs by diagnosing unbounded retry storms in the job orchestration layer and deploying a **Go/PostgreSQL** state-transition guard that enforced per-workflow retry limits, terminating infinite rework loops.
- Built a quality analytics dashboard in Retool backed by BigQuery materialized views for a **\$5M pilot**, surfacing per-user accept/reject rates, quality scores, and throughput for **2K+ annotators**, reducing ops reporting overhead by **85%**.
- Restored on-call signal quality by engineering an API middleware exclusion layer to filter expected errors pre-ingestion into Sentry, reducing alert noise by **55%** and incident **MTTR by 30%** across **100K+ daily requests**.

Machine Learning Engineering Intern

July 2024 – December 2024

- Designed a scalable LLM-driven Resume Parser, Expert Profiler, and Task Profiler pipeline processing **1M+ resumes** with **82% structured accuracy**, improving performance by **28%** over a rule-based baseline.
- Engineered an experience calculation algorithm for resume parsing and optimized the backend pipeline, reducing costs by **35%**.
- Operationalized an automated LLM-based PRP evaluation system, eliminating **90%** of manual review hours and reducing evaluation spend by **35%** across the annotation quality pipeline.
- Delivered a data ingestion pipeline on MongoDB and AWS S3 processing resume artifacts at scale, improving data retrieval latency by **45%**; ran analytical queries on a **3.6M-user, 120-table BigQuery** schema to surface active cohorts for model targeting.

Projects

Larynx Function Prediction | *Python, MATLAB* | **Published – Procedia Computer Science, Vol. 260, 2025**

- Published at RTIP2R-2024** (2000+ submissions); built an ML pipeline classifying larynx function from surface EMG signals using Tunable-Q Wavelet Transform feature extraction and W-KNN with 5-fold cross-validation.
- Surpassed **4 of 8 SOTA benchmarks** on classification task, validating the wavelet-based feature bank over conventional methods.

LLM-Powered Q&A System with SQL Integration | *Python, MySQL, LangChain, ChromaDB, Streamlit*

- Developed a Retrieval-Augmented Generation (RAG) system using LangChain and ChromaDB to translate natural language questions into executable SQL queries against a live MySQL database.
- Applied few-shot prompting and semantic embedding retrieval, achieving **92% NL-to-SQL** accuracy on a 500-query evaluation set; deployed as a Streamlit app with sub-second query response time.

Technical Skills

Languages & Frameworks: Golang, Python, JavaScript, TypeScript, C++, SQL, FastAPI, Gin, LangChain, gRPC, Streamlit.

Cloud & Infrastructure: AWS (S3, Amazon Neptune, CloudWatch), GCP (BigQuery), Docker, Kubernetes, CI/CD, RabbitMQ.

Databases & Storage: PostgreSQL, MongoDB, MySQL, ChromaDB, Redis.

AI/ML: LLMs, RAG, Vector Databases, BM25, Semantic Search, Prompt Engineering, Few-Shot Prompting, RLHF, NLP.

Systems & Architecture: Microservices, REST APIs, Modular Monolith, Observability (Sentry, Grafana, PGAnalyze, New Relic).

Leadership

Student Representative, Fest Review Committee – BITS Pilani

October 2023 – October 2024

- Selected 1/4 from 900+ by Campus Director to oversee **3Cr budget**; Spearheaded embezzlement prevention software saving **30L**.

Curator, TEDxBITSGoa

July 2022 – August 2023

- Led cross-functional **team of 55** to scale attendance by 5x and revenue 4x via data-driven marketing reaching **320K** people.